Industry adoption of IoT

According to Gartner, more than 30.9 billion IoT devices will be in use worldwide by 2025. A forecast by International Data Corporation (IDC) estimates that there will be 41.6 billion IoT devices in 2025, capable of generating 79.4 zettabytes (ZB) of data.

The global Internet of Things market is estimated to rise from US$ 478.36 billion in 2022 to US$ 2465.26 billion by 2029 at a CAGR of 26.4 per cent during the forecast period, as per *Global Wire*.

According to McKinsey, by 2030, the IoT market will be valued between US$ 5.5 trillion and US$ 12.6 trillion globally, including the value captured by consumers of IoT products and services.

Key characteristics of IoT analytics

IoT analytics integrates business intelligence and data analytics, including conventional and Big Data. Its characteristics are explained briefly below.

Volume: The volume of data processed by IoT data solutions is substantial and ever-growing. High data volumes demand massive data storage, data processing, data preparation, data curation, and data management processes. Big Data requires processing high volumes of data, that is, data of unknown value — for example, network traffic, sensor-enabled equipment capturing data, and many more. Some of the data sources that are responsible for generating high data volumes are:

- Scientific and research experiments
- Sensors, such as temperature, pressure and GPS sensors
- RFIDs
- Smart meters and telematics

Velocity: In IoT data analytics environments, massive data gets accumulated in a short time, as sensors collect data pretty quickly. The velocity of data arrival within an enterprise's perimeter is directly proportional to the time taken to process the data. It demands highly elastic, highly available data processing and massive data storage solutions for the enterprise.

Variety: IoT data is primarily unstructured or semi-structured. Such data types require additional processing to derive meaning and the supporting metadata. Further complexity arises when data from a known source changes without notice.

Veracity: Veracity refers to the quality or fidelity of data. Assessment of data that enters IoT data environments for quality can lead to data processing activities to resolve invalid data and remove noise. Data acquired in a controlled manner usually contains less noise than data acquired via uncontrolled sources. Data with a high signal-to-noise ratio has more veracity than data with a lower ratio.

Value: There is a range of quantitative and investigative techniques to derive value from data. The cost of data storage and computing has fallen exponentially, thus providing an abundance of data for statistical sampling and other techniques.

Response time and reliability: The essential requirement of IoT data analytics is better response time and reliability. The solution is built on three fundamental factors: deterministic analytics, data computation, and quick response time.

Bandwidth: The massive amount of data generated by IoT devices is gathered from control systems. Handling of this data requires better bandwidth. The increased network and infrastructure required to transfer data from one domain to another is compensated by creating valuable insights.

Security: Transferring raw data outside of controlled areas comes with security concerns and the associated costs. It may be more efficient to perform data analytics locally and share the necessary summaries with other domains.

Analytics maturity: Analytics involves the processing of raw data (measurements) into descriptions (information), and then contextualizing the results (knowledge) and benefiting from historical experience (wisdom). The maturity of the processed raw data depends on the design of the analytics solution.

Temporal correlation: Applying analytics closer to where the data is generated reduces the burden of correlation. It helps IoT systems to correlate the data between multiple sensors and process control system states.

Provenance: Performing analytics at a lower architecture tier maintains the genuine sources of the data as it progresses through the IoT layers.

Principles of IoT analytics

Architecture principles provide a basis for decision-making when developing IoT data analytics solutions and design. They form a structured set of ideas that collectively define and guide solution architecture development, harmonising decision-making across an organisation.

The principles that guide enterprises in the way they approach IoT analytics data are:

- *Data is an asset:* Data is an asset that has a specific and measurable value for the enterprise.
- *Data is shared:* Data must be shared across the enterprise and its business units. Users have access to the data that is necessary to perform their activities.
- *Data trustees:* Each data element has trustees accountable for data quality.
- *Common vocabulary and data definitions:* Data definition is consistent, and the taxonomy is understandable throughout the enterprise.
- *Data security:* Data must be protected from unauthorised users and disclosure.
- *Data privacy:* Privacy and data protection is considered throughout the life cycle of a Big Data project. All data sharing conforms to the relevant regulatory and business requirements.